Structured questions

|!|BCELB0060000001|!| ★ **NOS** **SI**

Before the discovery of vitamins, scientists had been suspecting that some foods (e.g. milk) contain certain factors essential to good health and humans require only small amounts of these factors.

The table below shows some investigations carried out by two scientists on the 'essential factors' in milk.

Scientist	Wilhelm Stepp (1882–1964)
Hypothesis	The essential factor(s) in milk might be associated with lipids.
Treatment	He mixed flour with milk to make 'milk bread'. He added alcohol and ether to the milk bread to extract the essential factor. **Experimental group:** The mice were fed with the treated milk bread only. **Control group:** The mice were fed with the treated milk bread and the extract.
Observation	**Experimental group:** All the mice died within three weeks. **Control group:** All survived normally after three weeks.

Scientist	Frederick Hopkins (1861–1947)
Hypothesis	Milk contains some 'accessory factors' which are essential for the healthy growth of young rats.
Treatment	He fed two groups of young rats with proteins, lipids, carbohydrates and minerals. He gave milk to one group only. After two weeks, he switched to feed the second group with milk.
Observation	When the mice were fed with milk, they grew normally. When the mice lacked milk, they did not grow well.

(a) Why did Stepp extract the essential factor in milk bread with alcohol and ether? (1 mark)

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(b) Did the results of Stepp's experiment support his hypothesis? (3 marks)

(c) What conclusion can be drawn from Hopkins' experiment? (1 mark)

(d) Elaborate on ***one*** aspect about the nature of science that can be demonstrated in the above events. (2 marks)

##

(a) He believed that the essential factor was associated with lipids, and that it would dissolve in organic solvents such as alcohol and ether. (1)

(b) Yes. (1)
The group fed with the treated milk bread only did not receive the essential factor and they died. (1)
The group fed with the extract survived normally. This shows that the essential factor dissolved in the alcohol and ether. (1)

(c) Milk contains some 'accessory factors' which are essential for the healthy growth of young rats. (1)

(d) Inquiry procedures are guided by the hypotheses made.
Stepp and Hopkins made different hypotheses. They treated the mice differently based on their own hypothesis. (1) + (1)
OR
Different investigative procedures may come to the same conclusions.
Stepp and Hopkins carried out different experiments, but they both reached the conclusion that milk contains some essential substances that are important for growth. (1) + (1)
OR
A good scientific experiment must include carefully designed controls.
Stepp and Hopkins both had set up experimental group and control group which differed only in the factor under investigation. (1) + (1)

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(Or other reasonable answers)

(7 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Analysing data	3
(c)	Practical skills	1
(d)	Applying concepts	2

|!|BCELB0060000002|!| ★ **STSE**

The table below shows some nutrition information about three types of milk.

	Whole milk (per 100 cm^3)	**Skimmed milk (per 100 cm^3)**	**High-calcium milk (per 100 cm^3)**
Protein	3.1 g	3.5 g	4.2 g
Total fat	3.5 g	0.3 g	1.5 g
Carbohydrate	4.5 g	5.3 g	6.5 g
Dietary fibre	0 g	0 g	0 g
Calcium	110 mg	170 mg	170 mg
Sodium	40 mg	57 mg	55 mg

(a) Which type of milk contains the highest energy value? (1 mark)

(b) Suggest ***two*** groups of people who would be recommended to drink skimmed milk instead of whole milk. (2 marks)

(c) Explain why skimmed milk is not suitable for babies. (2 marks)

(d) Suggest ***one*** group of people who would be recommended to drink high-calcium milk. (1 mark)

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(e) What health problems may arise if babies only consume whole milk for a long period of time? (2 marks)

_____

_____

##

(a) whole milk (1)

(b) obese / overweight people (1)

people with cardiovascular disease (1)

(c) Any ***one set*** of the following: (2)

- Skimmed milk has a low fat content. (1)
  Fat is important for forming cell structures / cell membranes in babies. (1)
- The energy value of skimmed milk is too low and (1)
  cannot meet the high metabolic demand of babies. (1)

(Or other reasonable answer and explanation)

(d) pregnant woman / elderly people (1)

(e) constipation (1)

scurvy (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Applying concepts	2
(c)	Analysing data	1
	Applying concepts	1
(d)	Applying concepts	1
(e)	Applying concepts	2

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|!|BCELB0060000003|!| ★ **STSE**

Jonathan consulted a nutritionist and was told that he has obesity. The doctor advised him to make changes to his diet and increase his daily energy expenditure. The table below shows the activities and the relative energy intake or consumption of Jonathan in a day.

Activity	Energy (kJ)
Breakfast	3500
Walk to school	2000
Lunch	4000
Do homework	3000
Play TV game	1500
Dinner	6000
Watch TV	1000

(a) Calculate the difference between energy input and energy output of Jonathan for one day. Explain why Jonathan has obesity. (5 marks)

(b) Jonathan searched a diet called ‘lean meats diet’, which encourages people to consume lean meats only in their diet. Jonathan thought that this diet could effectively reduce his weight.
If you were Jonathan’s doctor, explain the advantages and disadvantages of this diet. (4 marks)

##

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(a) Energy input = 3500 + 4000 + 6000 =13500 kJ (1)

Energy output = 2000 + 3000 + 1500 + 1000 = 7500 kJ (1)

Difference between energy input and output = 13500 – 7500 = 6000 kJ (1)

The energy intake of his diet is much greater than the energy output of the body, (1)

so the excess energy is stored as fats in the body. (1)

(b) Advantages: (Any ***two***, 2)

- Lean meats do not contain fats and carbohydrates, hence reducing his energy intake. (1)
- Lean meats are rich in proteins, which helps build muscles for his growth and development. (1)
- Lean meats contain a low level of cholesterol, which does not incur heart diseases or high blood pressure. (1)

(Or other reasonable answers)

Disadvantages: (Any ***two***, 2)

- Omission of essential nutrients, such as vitamins and minerals in the diet may incur health problems (1)
- Omission of dietary fiber in the diet may incur constipation. (1)

(Or other reasonable answers)

(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
	Mathematical skills	1
	Applying concepts	2
(b)	Applying concepts	4

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|!|BCELB0060000004|!| ★ **SI** **STSE**

In an investigation, 1 cm^3 of DCPIP solution was added into each of four test tubes W, X, Y and Z. A dropper was used to draw up a 0.1% vitamin C solution and was then added drop by drop to the DCPIP in tube W until the mixture changed its colour markedly.

This was repeated with the DCPIP solution in the other test tubes using the solutions shown in the table below. In each case the number of drops of solution required to react with 1 cm^3 of DCPIP was noted.

Tube	**Solution**	**Number of drops of solution needed to react with 1 cm^3 of DCPIP**
W	0.1 % vitamin C	15
X	Grape juice	28
Y	Lemon juice	5
Z	Boiled lemon juice	20

(a) State the colour change you would expect in tube W. (1 mark)

(b) In view of the results, arrange the four solutions in ascending order of vitamin C concentration. (1 mark)

(c) What conclusion would you draw from the results of tube Y and Z? Suggest an explanation for the conclusion. (2 marks)

(d) It was noted that after reacting with vitamin C, DCPIP became blue again if it was allowed to stand in air for some time. What does this observation suggest about the chemical nature of vitamin C? (1 mark)

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(e) Suggest and explain ***two*** ways by which vitamin C in fresh vegetables can be preserved. (4 marks)

##

(a) From deep blue to colourless (1)

(b) X < Z < W < Y (1)

(c) Boiled lemon juice contained less vitamin C than lemon juice. (1)
Vitamin C was destroyed by heating. (1)

(d) Reducing / anti-oxidant in nature (1)

(e) Store the vegetables in a refrigerator (at 4 ^{o}C) (1)
so as to prevent the destruction of vitamin C by high temperatures. (1)
Not to over-wash / soak the vegetables in water for a long time (1)
so as to prevent vitamin C from dissolving away. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	1
(c)	Practical skills	2
(d)	Applying concepts	1
(e)	Applying concepts	4

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|!|BCELB0060000005|!| ★ **SI**

Tom conducted various food tests to investigate the chemical compositions of four natural foodstuffs. The results are summarized in the table below.

Foodstuffs	Benedict's test	Iodine test	Grease spot test	Protein test strip	Test X
A	blue	brown	no translucent spot left	negative	deep blue
B	orange	brown	translucent spot left	positive	colourless
C	orange	blue-black	no translucent spot left	positive	colourless
D	blue	brown	translucent spot left	negative	deep blue

(a) Foodstuffs A to D are all solid foods. Outline the steps of preparing extracts from solid foods. (3 marks)

(b) In view of the results of Test X, suggest what Test X is. (1 mark)

(c) Using the letters in the table, identify the foodstuff(s) which

(i) contain(s) reducing sugar and starch. (1 mark)

(ii) do(es) not contain lipids but contain(s) proteins. (1 mark)

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(d) According to the results of foodstuff C, state whether it is a food of plant or animal origin. Cite and explain one piece of evidence to support your answer. (3 marks)

(e) In view of your answer to (d), suggest a food substance not revealed in this experiment which is most likely to be present in foodstuff C. Name the health problem arising from the deficiency of this food substance in the diet. (2 marks)

##

(a) Cut the food sample into small pieces. (1)
Grind the food pieces in a mortar with distilled water. (1)
Filter the ground food sample by squeezing it through three layers of moistened muslin cloth. Collect the food extract. (1)
(b) DCPIP test (1)
(c) (i) C (1)
(ii) C (1)
(d) Foodstuff C is a food of plant origin. (1)
It contains starch (1)
which is only found in plant tissues. (1)
(e) dietary fibre (1)
constipation (1)

(11 marks)

##

	Skills assessed	Marks
(a)	Practical skills	3
(b)	Analysing data	1
(c)(i)	Analysing data	1
(c)(ii)	Analysing data	1
(d)	Analysing data	1
	Applying concepts	2
(e)	Understanding basic concepts	2

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|!|BCELB0060000006|!| ★ **STSE**

Yvonne is very concerned about her body weight. She follows a diet found on the Internet in order to lose weight. The diet contains a large amount of proteins. Someone claims that proteins can satisfy hunger better than either fats or carbohydrates. The diet and its nutrition values (per 100 g) are shown in the table below.

		Carbohydrates (g)	**Fat (g)**	**Protein (g)**	**Dietary fibre (g)**
Breakfast	Egg	0.8	9.9	12.5	0
	Tomato	3.9	0.2	0.9	1.2
	Milk	4.99	0.97	3.37	0.0
Lunch	Chicken breast	0	1.6	22.3	0
	Lettuce	5.8	0.1	1.3	2.5
	Coffee	0.1	0	0.1	0
Dinner	Beef	0	5.79	33.5	0
	Pork	0	4.06	22.39	0
	Cauliflower	3.6	0.22	1.6	2.7

(a) How does the human body metabolize proteins to provide energy for daily activities? (3 marks)

(b) If Yvonne follows this diet for three months, her liver may be damaged. Explain why. (1 mark)

(c) Suggest and explain another health problem that may be caused by adopting this diet. (2 marks)

(d) The diet shown above contains a little amount of carbohydrates.

(i) How does this diet affect the blood glucose level? (1 mark)

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(ii) What is the possible effect of the change in the blood glucose level stated in (i) to the body? (2 marks)

_____

_____

##

(a) Proteins are broken down into amino acids (in the digestive tract). (1)
Excess amino acids are deaminated to form carbohydrates or lipids (1)
which can be used for respiration to release energy. (1)

(b) This diet contains a large amount of proteins. The liver has extra workload to deaminate the excess amino acids. (1)

(c) Any ***one set*** of the following: (2)

- Heart disease (1)
  because the diet includes a lot of red meat and fat, which may cause the blockage of blood vessels (1)
- Kidney problems (1)
  because the increased breakdown of amino acids produce a large amount of urea, which may increase the workload of the kidneys (1)
- Constipation (1)
  because the diet contains very little dietary fibre (1)

(Or other reasonable answer and explanation)

(d) (i) The blood glucose level will become low. (1)

(ii) When the blood glucose level is low, brain cells may not be supplied with enough glucose for respiration to release energy. (1)
The person may faint. (1)
(Or other reasonable answers)

_____
(9 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	3
(b)	Applying concepts	1
(c)	Analysing data	2
(d)(i)	Applying concepts	1
(d)(ii)	Applying concepts	2

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|!|BCELB0060000007|!| ★ **STSE**

The table below shows the amount of different food substances in a cereal product.

	Amount per 100 g of cereals
Protein	10.3 g
Carbohydrates	63.8 g
Lipid	2.9 g
Dietary fibre	15.3 g
Iron	17.6 mg
Salt	0.8 g

(a) According to the table above, explain why this cereal product is suitable for an 80-year-old man. Give ***two*** reasons. (4 marks)

(b) Given that the energy provided by 1 g of protein, carbohydrate and lipid is 17 kJ, 18 kJ and 38 kJ respectively. Calculate the energy value of 100 g of this cereal product. (2 marks)

(c) The recommended daily energy intake for an 80-year-old man is 8300 kJ. If the breakfast is to account for 35% of the elderly daily energy intake, how many grams of the cereal product should he eat? (2 marks)

##

(a) Any ***two sets*** of the following: (4)

- It is low in lipid (1)
  which can prevent heart disease / obesity / overweight. (1)
- It is high in dietary fibre (1)
  which helps prevent constipation. (1)

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- It is low in salt content (1)
  which reduces the risk of high blood pressure. (1)

(b) Energy value of 100 g of the cereal product
= (10.3 × 17 + 63.8 × 18 + 2.9 × 38) kJ (1)
= 1433.7 kJ (1)

(c) Energy to be gained in breakfast = 8300 × 0.35 = 2905 kJ (1)
Weight of cereals required = (2905/1433.7) × 100 g = 203 g (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
	Applying concepts	2
(b)	Mathematical skills	2
(c)	Mathematical skills	2

|!|BCELB0060000008|!| ★

Many fruits are rich in vitamin C. The vitamin C content of fruit often decreases during storage.

(a) A freshly-picked orange is found to contain 100 mg of vitamin C. Another similar-sized orange was stored for two weeks and is found to contain 73 mg of vitamin C.
Calculate the percentage decrease in vitamin C content of orange after storage of two weeks. (2 marks)

(b) (i) State ***two*** properties of vitamin C which lead to the loss of vitamin C during the storage of fruits. (2 marks)

(ii) According to the properties of vitamin C, suggest ***two*** ways to reduce the loss of vitamin C during storage. (2 marks)

(c) (i) What is the deficiency disease of vitamin C? (1 mark)

_____

(ii) Explain how a lack of vitamin C causes this disease. (2 marks)

_____

_____

##

(a) Decrease in vitamin C content = 73 mg – 100 mg = –27 mg (1)

Percentage decrease in vitamin C content = (–27 mg / 100 mg) × 100% = –27% (1)

(b) (i) Vitamin C is easily oxidized by oxygen in the air. (1)

Vitamin C is destroyed at higher temperatures. (1)

(ii) Any ***two*** of the following: (2)

- storage with low oxygen
- coating with wax (to reduce oxygen entry)
- keeping in a cool environment

(Or other reasonable answer)

(c) (i) scurvy (1)

(ii) Vitamin C is needed in the formation of connective tissues. (1)

A lack of vitamin C will lead to bleeding gums / poor healing of wounds. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Mathematical skills	2
(b)(i)	Applying concepts	2
(b)(ii)	Applying concepts	2
(c)(i)	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	2

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|!|BCELB0060000009|!| ★ **STSE**

Read the passage below and answer the questions that follow.

> Obsessive slimming was especially common in the modelling industry. Extreme slimness among models has become a controversy since two young models died from anorexia nervosa in 2006. One of the models, Reston, was 168 cm tall and weighed 40 kg when she died. Reston lived on a diet of apples and tomatoes before her death.
>
> The deaths of the models led to a move by Madrid government to ban models with a BMI less than 18 from performing in fashion shows. Italian designers have also agreed to demand all models to submit medical reports as proof that they do not suffer from an eating disorder.

(a) Based on the given information, calculate Reston's BMI. (2 marks)

(b) Explain whether or not Reston has a balanced diet. (3 marks)

(c) What health problems would anorexia nervosa lead to? Give ***two*** examples. (2 marks)

(d) Suggest ***one*** treatment for anorexia nervosa. (1 mark)

##

(a) Reston's BMI = $40 / (1.68)^2$ (1)
= 14.17 (1)

(b) Reston did not have a balanced diet. (1)
She had a diet of apples and tomatoes which are rich in carbohydrates, water and vitamin C, (1)
but insufficient in lipids and proteins. (1)

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(c) Any ***two*** of the following: (2)

- osteoporosis
- kidney damage
- heart disease

(Or other reasonable answers)

(d) psychotherapy / drug treatment / other reasonable answer (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Mathematical skills	2
(b)	Applying concepts	3
(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	1

|!|BCELB0060000010|!| ★ **STSE**

The following table shows the average amount of food intake by Martin per day and a diet recommended by a nutritionist.

Foodstuff	Average intake of foodstuff (g)	
	Martin's diet	Recommended diet
Cereals	83	85
Meat	60	40
Fish	10	7
Diary products	27	25
Fat and oil	35	20
Potatoes	60	100
Sugar	42	40
Vegetables	24	70
Fruits	30	65

(a) Which foodstuffs are the main sources of starch in Martin's diet? (1 mark)

_____

(b) Deduce which ***two*** food substances are taken in excess by Martin. (1 mark)

_____

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(c) What would happen to the two food substances mentioned in (b) if they are consumed in excess? (4 marks)

(d) Identify from the table a food substance that Martin lacks. Suggest a consequence of lacking this food substance. (2 marks)

##

(a) cereals and potatoes (1)
(b) lipid and protein (1)
(c) Excess lipid is stored in the subcutaneous fat (1)
or forms adipose tissue around internal organs (e.g. kidneys and heart). (1)
Proteins are digested to amino acids. (1)
Excess amino acids are then broken down by deamination in the liver. (1)
(d) dietary fibre (1)
He may suffer from constipation. (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	1
(c)	Understanding basic concepts	4
(d)	Analysing data	1
	Understanding basic concepts	1

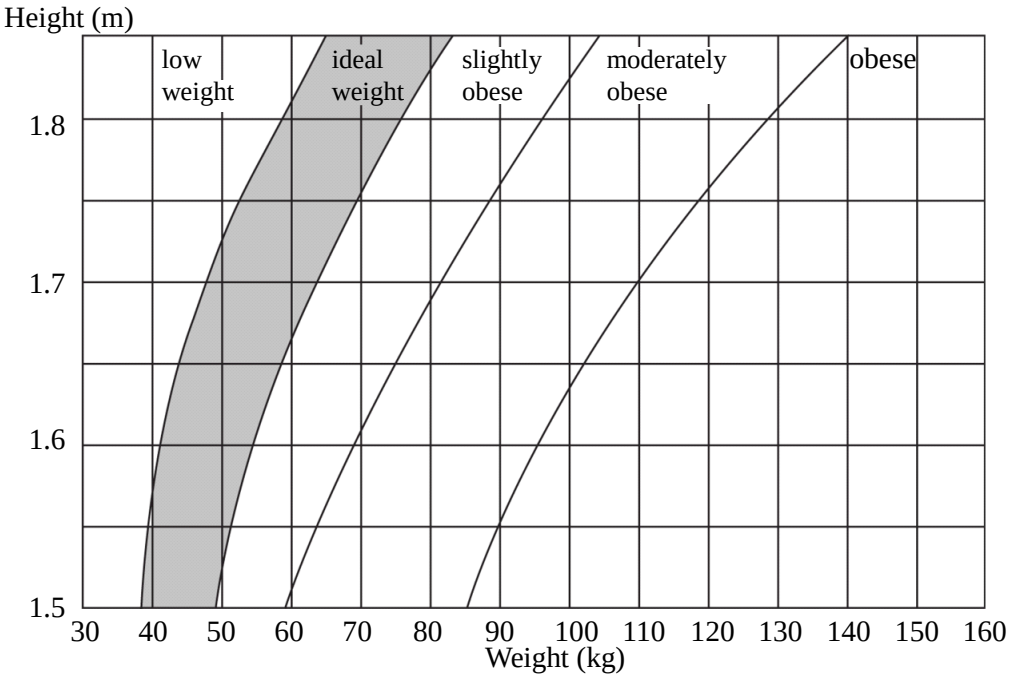
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|!|BCELB0060000011|!| ★ **STSE**

In a body check-up, Robert, George and Jimmy have their weights and heights measured.

	Weight (kg)	**Height (m)**
Robert	50	1.55
George	50.1	1.75
Jimmy	90	1.65

The chart below can be used to find out the degree of obesity of a person, such as 'low weight', 'ideal weight', 'slightly obese', 'moderately obese' and 'obese.'



(a) According to the chart, which category do Robert, George and Jimmy fall in?

(3 marks)

(b) Obesity increases the risk of certain health problems. Suggest ***two*** of the problems.

(2 marks)

(c) Which ***two*** food substances, when eaten in excess, would likely lead to obesity? (2 marks)

_____

_____

(d) Suggest ***two*** ways to prevent obesity. (2 marks)

_____

_____

##

(a) Robert: ideal weight (1)
George: low weight (1)
Jimmy: moderately obese (1)

(b) Any ***two*** of the following: (2)

- heart disease
- high blood pressure
- diabetes mellitus

(Or other reasonable answers)

(c) lipids (1)
carbohydrates (1)

(d) Any ***two*** of the following: (2)

- have a balanced diet
- exercise regularly

(Or other reasonable answers)

(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	3
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	2

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|!|BCELB0060000012|!| ★ **STSE**

Rose analysed the nutrition composition of tomatoes, potatoes, peas and onions and recorded in the table below.

Vegetable (100 g)	**Total carbohydrate (g /100 g)**	**Starch (g /100 g)**	**Dietary fibre (g /100 g)**
Tomatoes	3	Trace	1
Potatoes	17	16	1
Peas	9	5	5
Onions	4	Trace	1

(a) The amount of total carbohydrate is calculated as the sum of the amounts of sugars and starch. With reference to the table, which vegetable has the highest content of sugars? (1 mark)

_____

(b) There are 17 mg of vitamin C in 100 g of peas. If Rose eats 100 g of peas, she obtains 25% of the recommended daily allowance (RDA) of vitamin C.

(i) Calculate the RDA of vitamin C for Rose. Show your working. (2 marks)

_____

_____

(ii) State the disease caused by the deficiency of vitamin C. (1 mark)

_____

(c) In some developing countries, many people suffer from prolonged starvation.

(i) If you are from a charity organization, which of the four foods from the above table will you bring to the people in developing countries? Explain your choice. (2 marks)

_____

_____

(ii) People in countries with famine may have protein deficiency. Name the disease caused by protein deficiency and describe an effect of this disease on children's growth. (2 marks)

_____

_____

##

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(a) peas (1)

(b) (i) RDA of vitamin C for Rose = 17 mg / 25% (1)
= 68 mg (1)

(ii) scurvy (1)

(c) (i) potato (1)
It contains the greatest amount of carbohydrates which are the main source of energy. (1)

(ii) kwashiorkor (1)
stunted growth / low muscle mass / hair loss / swelling in the abdomen / other reasonable answer (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)(i)	Mathematical skills	2
(b)(ii)	Understanding basic concepts	1
(c)(i)	Analysing data	1
	Applying concepts	1
(c)(ii)	Understanding basic concepts	2

|!|BCELB0060000013|!| ★

Vitamins and minerals are important to our health.

(a) Give ***two*** similarities and ***two*** differences between vitamins and minerals. (4 marks)

(b) Which vitamin and mineral are essential for bone formation? (2 marks)

(c) Which pigment are carrots rich in? State the importance of this pigment to the human body. (3 marks)

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(d) Anaemia is a disease of iron deficiency. The symptoms of anaemia include feeling tired easily and fainting. Explain why the deficiency of iron would lead to these symptoms. (3 marks)

##

(a) Similarities:

Both provide no energy to the human body. (1)

Both are needed in very small amounts for maintaining good health. (1)

(Or other reasonable answer)

Differences: (Any ***two***, 2)

- Vitamins are organic substances while minerals are inorganic.
- Some types of vitamins (e.g. vitamin D) can be synthesized by the human body, but minerals cannot be synthesized by the human body.
- Some types of vitamins are water-soluble and some are lipid-soluble. Minerals are not soluble in lipids.

(Or other reasonable answer)

(b) vitamin D (1)

calcium (1)

(c) carotene (1)

Carotene can be converted to vitamin A. (1)

Vitamin A is needed for the formation of visual purple, the pigment responsible for night vision. (1)

(d) Iron is a component of haemoglobin, (1)

which is the oxygen-carrying pigment in red blood cells. (1)

A lack of iron would lead to inadequate oxygen supply to body parts, including the brain. (1)

(12 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	4
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	3

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(d)	Applying concepts	3

|!|BCELC0060000014|!|

In an experiment, the contents of three test tubes (X, Y and Z) were analysed. Each test tube contains at least one type of food substance. Food tests were carried out and the results are shown in the following table.

	X	**Y**	**Z**
Iodine test	+	–	–
Grease spot test	–	–	+
Using protein test strips	+	+	–
Benedict's test	–	+	–

Key: + positive result – negative result

(a) Which of the four food tests requires heating? (1 mark)

(b) Which food substances are present in tubes X, Y and Z, respectively? (3 marks)

(c) Describe the positive results of the four food tests. (4 marks)

##

(a) Benedict's test (1)

(b) Tube X contains starch and protein. (1)
Tube Y contains protein and reducing sugars. (1)
Tube Z contains lipids. (1)

(c) Iodine test: Iodine solution turns from brown to blue-black. (1)
Grease spot test: A translucent spot is formed on the filter paper, which persists unless immersed in an organic solvent. (1)
Protein test strip: A distinctive colour change is observed. (1)
Benedict's test: A brick-red precipitate appears. (1)

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(8 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Analysing data	3
(c)	Understanding basic concepts	4

|!|BCELB0060000015|!| ★

The table below gives the energy value (kJ/g) and composition of some common foodstuffs.

Food	Energy (kJ/g)	% W	% X	% Y	% Z
Potatoes	3.3	15	--	2	82
Fish	4.8	--	13	16	71
Milk	2.7	4.5	3.5	3.4	88.2
Butter	54	--	83	0.6	16.4
Beef	21.8	--	21	24	54
Soya beans	3.2	28	20	32	15
Maize	3.5	55	11	14	20

(a) Which food in the table has

(i) the highest energy value? (1 mark)

(ii) the lowest energy value? (1 mark)

(b) W, X, Y and Z represent lipids, proteins, carbohydrates or water. Identify the food substance represented by each letter. (4 marks)

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(c) State ***two*** other food substances which are essential for health, but are absent in the table. Describe their functions. (4 marks)

(d) Vegetarians are advised to eat a wide range of plant foods including beans and peas. With reference to the table, explain this advice. (2 marks)

##

(a) (i) butter (1)
(ii) milk (1)
(b) W: carbohydrates (1)
X: lipids (1)
Y: proteins (1)
Z: water (1)
(c) Any ***two sets*** of the following: (4)
- minerals (1)
  regulate metabolic reactions / build body tissue (1)
- vitamins (1)
  regulate metabolic reactions in the body (1)
- dietary fibre (1)
  stimulates peristalsis of the alimentary canal (1)

(d) Beans contain a high proportion of proteins. (1)
Proteins are important for growth and repair of tissues. (1)

(12 marks)

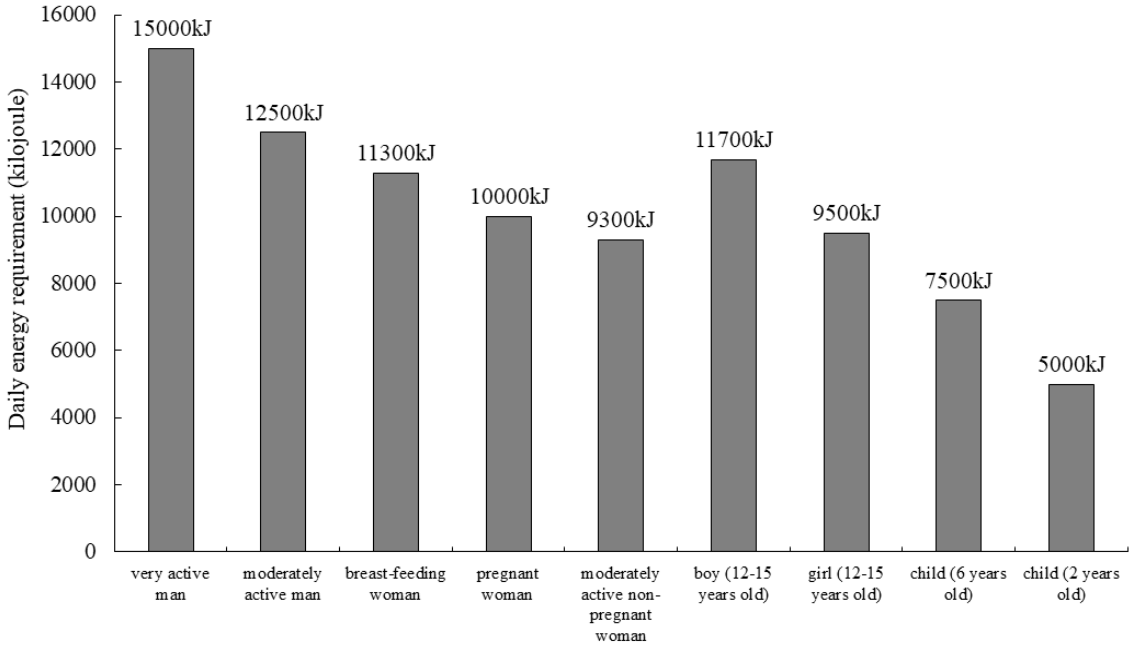
##

	Skills assessed	Marks
(a)(i)	Analysing data	1
(a)(ii)	Analysing data	1
(b)	Analysing data	4
(c)	Understanding basic concepts	4
(d)	Analysing data	1
	Applying concepts	1

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|!|BCELB0060000016|!| ★

The graph below shows the daily energy requirements of different groups of people.



(a) What is the difference in the energy requirement between males and female who are both moderately active? Explain why. (2 marks)

(b) What is the difference in the energy requirement between a pregnant woman and a non-pregnant woman? Explain the difference. (2 marks)

(c) Why do a woman's calcium and vitamin D requirements increase when she is

(i) pregnant? (2 marks)

(ii) breast-feeding her baby? (2 marks)

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##

(a) Males have a higher energy requirement than females because (1)
they tend to have a larger body size and have more muscle tissues than females / males lose heat faster than females due to a lower percentage of body fat. (1)

(b) A pregnant woman has a higher energy requirement than a non-pregnant woman (1)
because a pregnant woman has to supply nutrients for the development of her foetus. (1)

(c) (i) Calcium and vitamin D are very important for the development of bones of the foetus in a pregnant woman. (1)
The woman herself also requires calcium and vitamin D for maintaining the calcium level in her bones. (1)

(ii) The woman's body has to produce breast milk (1)
which can supply calcium and vitamin D to her baby for development of bone and teeth. (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
	Understanding basic concepts	1
(b)	Analysing data	1
	Understanding basic concepts	1
(c)(i)	Understanding basic concepts	2
(c)(ii)	Understanding basic concepts	2

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|!|BCELB0060000017|!| ★

Ellen is an office lady weighing 61 kg and 158 cm tall. **Table 1** lists the diet that Ellen has consumed for years. However, her doctor now advises her to change her eating habit and gives her a suggested diet (**Table 2**). **Table 3** and **Table 4** show the calculation of BMI and the daily energy requirement of different people.

Table 1

	Foodstuff	Amount	Energy (kJ)
Breakfast	ham	2 pieces	340
	toast (with butter)	1 piece	830
	fried egg	2	670
	fried pork chop	1 piece	1510
	hot milk tea	1 cup	440
Lunch	roast pork and rice	1 dish	2520
	coke	1 can	630
Dinner	seafood soup	1 bowl	1890
	meat sauce spaghetti	1 dish	3150

Table 2

	Foodstuff	Amount	Energy (kJ)
Breakfast	ham sandwich	1	1340
	skim milk	1 glass	340
Lunch	rice noodles with fishballs	1 bowl	1340
	boiled vegetables	1 dish	190
	apple	1	190
Dinner	steamed fish	1 dish	1050
	rice	1 bowl	1050
	boiled vegetables	1 dish	190
	orange	1	190

Table 3

Body mass index (BMI) formula	**BMI range**	
$\text{BMI} = \frac{\text{Weight (kg)}}{[\text{Height (m)}]^2}$	Underweight	< 18.5
	Ideal	$18.5 - 23$
	Overweight	$23 - 24.9$
	Obese	> 24.9

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Table 4

Activity level	Daily energy requirement (kJ/kg)	Occupation
Low	130	office workers, salespersons
Medium	150	housewives, nurses
High	170	construction workers

(a) According to **Table 3**, calculate the body mass index (BMI) of Ellen. Which range does her BMI fall into? (2 marks)

(b) What disease will Ellen probably suffer from if she continues to have her present diet? Explain your answer in terms of energy intake and output, with reference to the given data. (4 marks)

(c) State ***one*** food substance that is severely inadequate in her present diet. Give ***one*** importance of this food substance to the human body. (2 marks)

##

(a) Ellen's BMI = 61 / $(1.58)^2$
= 24.44 (1)
Her BMI falls in the range of 'Overweight'. (1)

(b) obesity (1)
Ellen's current energy intake = (340 + 830 + 670 + 1510 + 440 + 2520 + 630 + 1890 + 3150) kJ
= 11980 kJ (1)
Ellen's daily energy requirement = 61 kg ×130 kJ/kg
= 7930 kJ (1)
The net intake of energy per day = 11980 kJ – 7930 kJ

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= 4050 kJ (1)

The net intake is so large that she will soon have obesity if she continues to have this diet.

(c) Any ***one set*** of the following: (2)

- dietary fibre (1)
  it can stimulate peristalsis of alimentary canal / prevents constipation (1)
- vitamin C (1)
  it helps in the formation of connective tissue and healing of wounds (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Mathematical skills	1
	Analysing data	1
(b)	Analysing data	1
	Mathematical skills	3
(c)	Analysing data	1
	Understanding basic concepts	1

|!|BCELB0060000018|!| ★

Table 1 shows the daily nutritional requirements of an infant (less than one year old) and an adult, while **Table 2** shows the main ingredients of cow's milk (per kg).

Table 1	**Daily energy requirement (kJ)**	**Protein (g)**	**Calcium (mg)**	**Iron (mg)**	**Vitamin D (μg)**
Infant	3000	14	600	6	10
Adult	12500	42	500	10	2.5

Table 2	**Energy content (kJ)**	**Protein (g)**	**Calcium (mg)**	**Iron (mg)**	**Vitamin D (μg)**
Cow's milk	2700	32	1200	1	0.3 (summer) 0.1 (winter)

(a) State the function of iron in an infant who is less than one year old. (1 mark)

(b) What is the importance of protein to an adult? (1 mark)

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(c) Explain why an infant needs more calcium and vitamin D than an adult. (1 mark)

(d) Explain why cow's milk contains more vitamin D in summer than in winter. (1 mark)

(e) Calculate how much cow's milk would an infant need per day in order to acquire sufficient protein. (2 marks)

(f) From the table, which food substance in cow's milk is insufficient for the growth of the infant? How can the infant obtain the food substance that is insufficient in cow's milk? (2 marks)

##

(a) for formation of haemoglobin (1)

(b) for making new cells to replace worn out tissues / repair body tissues (1)

(c) for formation of bones and teeth (1)

(d) In summer, more sunlight is available. Thus a cow can form more vitamin D in its skin. (1)

(e) Amount of cow's milk required by an infant per day = 14/32 kg (1)
= 0.4375 kg (1)

(f) Any ***one set*** of the following: (2)

- Iron (1)
  ingest eggs / other iron-containing food that is suitable for infants (1)
- Vitamin D (1)
  ingest fish-liver oil / get enough sun exposure (so that the skin synthesizes enough vitamin D) (1)

(8 marks)

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Applying concepts	1
(e)	Mathematical skills	2
(f)	Analysing data	1
	Understanding basic concepts	1

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